

GCP IAM Security Lab Report

Implementing Least-Privilege Access Control Using Service Accounts and Custom IAM Roles in Google Cloud Platform

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Role

Future Cloud Security Engineer

Project Type

Cloud Security Engineering Lab

Platform

Google Cloud Platform (GCP)

Security Domain

Identity and Access Management (IAM)

Report Date

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Executive Summary

Identity and Access Management (IAM) is one of the most critical security controls within cloud environments. Misconfigured permissions, excessive privileges, and unmanaged identities remain among the leading causes of cloud security incidents.

This project demonstrates the implementation of secure IAM controls within Google Cloud Platform (GCP) using service accounts, predefined roles, custom IAM roles, and policy verification through the Google Cloud Console and gcloud CLI.

The primary objective was to design and validate a least-privilege access model that minimizes unnecessary permissions while maintaining operational functionality. The project successfully established role-based

access control (RBAC), created a dedicated service account, implemented a custom IAM role, and validated permissions using command-line policy inspection.

The project follows cloud security best practices and reflects real-world IAM administration tasks commonly performed by Cloud Security Engineers.

Project Objectives

The objectives of this project were:

- Create and configure a Google Cloud project
 - Enable required cloud services
 - Configure Identity and Access Management controls
 - Create and secure service accounts
 - Implement least-privilege permissions
 - Develop a custom IAM role
 - Assign and validate IAM policies
 - Demonstrate IAM automation using gcloud CLI
 - Apply cloud governance and access control best practices
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Lab Environment

Component	Description
Cloud Platform	Google Cloud Platform (GCP)
Security Service	IAM & Admin
Authentication	Google Account
Access Control Model	RBAC
Service Identity	Service Account
Management Tool	Google Cloud Console
Automation Tool	gcloud CLI
API Enabled	Compute Engine API

Security Architecture

The project follows a role-based access control architecture.

User Account
↓
IAM Policy
↓
Service Account
↓
Custom IAM Role
↓
Compute Engine Permissions

This architecture ensures permissions are assigned through controlled IAM policies rather than unrestricted administrative access.

Implementation Process

Phase 1: Project Initialization

A dedicated Google Cloud project named:

gcp-iam-security-lab

was created to provide an isolated environment for security testing and IAM configuration.

Activities Performed

- Created GCP project
- Configured project settings
- Verified project ownership

Security Value

Using a dedicated project prevents unauthorized access to production resources and enables controlled testing.

Phase 2: Billing and API Configuration

Billing was linked to the project and required services were enabled.

Enabled Service

Compute Engine API

Security Value

Only required APIs were enabled, reducing unnecessary exposure and limiting the potential attack surface.

Phase 3: Service Account Creation

A dedicated service account was created.

Service Account

cloud-security-viewer

Purpose

The service account represents non-human identities used by applications and automation workflows.

Security Controls Applied

- Dedicated service identity
- Minimal permissions
- No service account keys generated

Security Benefit

Eliminates excessive user-based permissions and improves accountability.

Phase 4: Role Assignment

The Viewer role was assigned to the service account.

Assigned Role

Viewer

Capabilities

- View project resources
- Inspect configurations
- Verify permissions

Security Benefit

Provides read-only access while preventing resource modification.

Phase 5: Custom IAM Role Development

A custom IAM role was created to demonstrate granular permission management.

Custom Role Name

CustomRole

Included Permissions

- compute.instances.list
- compute.instances.start
- compute.instances.stop

Implementation Method

Custom role created using role-definition.json and deployed through gcloud CLI.

Security Benefit

Reduces privilege escalation risk by limiting permissions to required actions only.

IAM Policy Assignment

The custom role was assigned to the service account using IAM policy bindings.

Assignment Method

gcloud projects add-iam-policy-binding

Result

The service account successfully received:

- Viewer Role
- CustomRole

Security Benefit

Demonstrates controlled permission delegation using policy-based access management.

Access Verification

IAM policies were verified using command-line inspection.

Verification Command

```
gcloud projects get-iam-policy
```

Validation Results

Confirmed:

- Viewer Role assigned
- Custom IAM Role assigned
- Policy binding successfully applied

Evidence Collected

- Service Account Creation Screenshot
 - Custom Role Creation Screenshot
 - Custom Role Assignment Screenshot
 - IAM Verification Screenshot
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Security Best Practices Implemented

Principle of Least Privilege

Only required permissions were granted.

Role-Based Access Control

Access managed through predefined and custom roles.

Service Account Security

Dedicated service account used instead of direct user permissions.

Keyless Authentication

No service account keys created.

IAM Verification

Policies validated using CLI.

Cloud Governance

Access centrally managed through IAM.

Risk Analysis

Risk	Mitigation
Excessive Permissions	Custom IAM Role
Unauthorized Resource Modification	Viewer Role
Credential Exposure	No Service Account Keys
Misconfigured Access	Policy Verification
Privilege Escalation	Least Privilege Model

Cloud Security Skills Demonstrated

This project demonstrates practical experience in:

- Cloud Identity and Access Management
 - Google Cloud Security
 - IAM Administration
 - Service Account Management
 - Role-Based Access Control
 - Principle of Least Privilege
 - Cloud Governance
 - Security Automation
 - gcloud CLI Operations
 - Security Documentation
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Business Impact

Proper IAM configuration is essential for securing cloud infrastructure.

Benefits achieved include:

- Reduced attack surface
 - Improved access control
 - Enhanced governance
 - Better compliance readiness
 - Stronger cloud security posture
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Cleanup Activities

Following project completion:

- Service account removed
- Custom role removed
- IAM policy bindings removed
- Project deleted

This ensured no residual resources remained active and prevented unnecessary cloud costs.

Conclusion

This project successfully implemented a secure Identity and Access Management architecture within Google Cloud Platform using industry-standard security practices.

The project demonstrated the creation and management of service accounts, implementation of least-privilege permissions, development of custom IAM roles, and validation of IAM policies through automation.

These activities reflect real-world cloud security responsibilities commonly performed by Cloud Security Engineers and IAM Administrators, providing practical hands-on experience in securing cloud identities and access controls.

The successful completion of this project strengthened foundational skills in cloud governance, access management, and security automation while reinforcing best practices for protecting cloud environments.